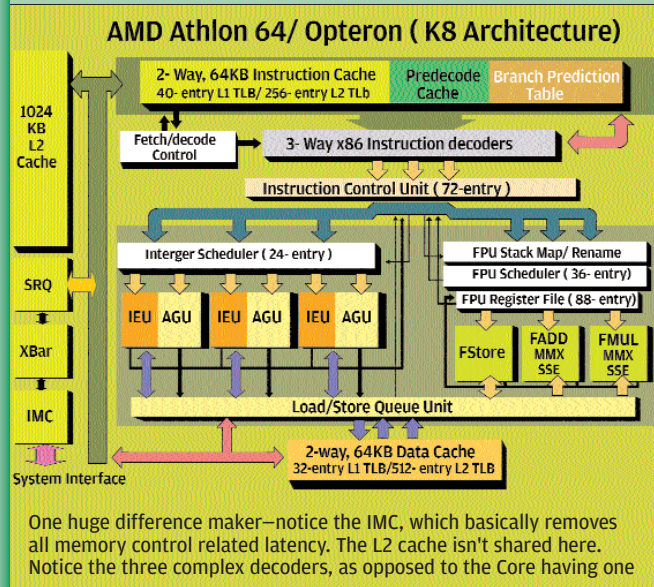


KB for each core); that's 32 KB for data and 32 KB for instructions. Earlier, Pentium D processors had exactly half the L1 cache.

What Intel did with the Conroe's L2 cache is somewhat revolutionary—it's shared! The huge 4 MB of L2 cache is completely shared between the two cores.

Let's take a hypothetical situation. If one of the cores needs more cache during operation (assuming the other core to be idle), all the 4 MB is at its disposal. Intel has termed this technology "Advanced Smart Cache."

Let's look at another scenario, where Advanced Smart Cache scores big time! Suppose both the cores are independently working on the same data synchronously, the data needs be stored only once in the cache, and not twice as in case of traditional discrete-cache CPUs. Normally, each of the cores would need a copy of this data in their respective L2 caches, causing redundancy, and therefore wastage of cache. This is a huge advantage for the Core architecture over Intel's previous products, and AMD's current products. Furthermore, each core would have to communicate with the other to ensure that the other hasn't modified the data in its cache in any way, since both caches need the latest



Out of Order Execution means exactly what the phrase suggests. An instruction needs to be executed immediately, and therefore gets preference over other pipelined and scheduled instructions, the Core 2 Duo excels at Out of Order execution operations

copies of the data, else "Cache Contamination" will occur. This communication causes CPU overhead, and since it must occur via the FSB, the data bus is further bogged down.

Wide Dynamic Execution Revisited

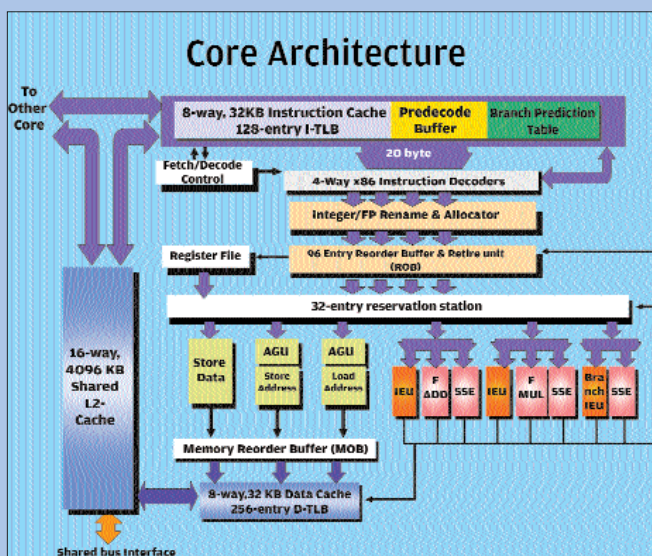
One of the technologies present in the Conroe touted by Intel isn't new at all: "Wide Dynamic Execution" has been at work since Pentium Pro days! What "wide" means is a higher IPC (Instructions Per Clock) or commands per clock count. While each of the cores on the Conroe can handle a minimum of four instructions per clock, previous-generation Pentium 4s could handle a maximum of three. The Core architecture

which can be, very simply, of two types—opcode and addresses, or operations and locations. The decoder on a CPU is tasked with deciphering these instructions and reducing the variance of their length—forming even or close-to-even sized packets, since x86 instructions can be between one and 15 bytes long. In the bargain, RISC-like instructions are created, which make the job of scheduling and execution that much easier.

The Core architecture features three simple and one complex decoder. AMD's K8 features three complex decoders, without any simple decoder.

Typically, the simple decoders are handed the task of working on the most-frequently-used x86 instructions, which are converted into micro-ops, typically a single micro-op per set of instructions. The complex decoders work on more of the heavier CISC-like instructions, and will produce more than one micro-op per operation. Along with the AMD 64 CPUs, older Athlon XPs and Intel's Pentium 4 and Pentium III CPUs all use this method of decoding.

But AMD 64s have 3 complex decoders. How do they stack up? There are two ways of decoding: a direct, easier path, and a vector path that is more suitable for certain complex instructions, but which may take longer to decode. AMD 64s use both these ways of decoding instructions. Each of the complex decoders can perform both direct and vector path decoding, but direct path decoding is preferred because the resultant macro-ops are fewer in number. When it comes to decoding complex instructions, the K8 architecture would definitely score big time. However, x86 instructions are largely simple,



Note the 4 simple decoders. The reservation station improve data bandwidth. Note the massive L2 cache shared between cores, Also notice the larger entry Translation Look-Ahead Buffer in the L1 cache

It's All About The Cache!

AMD 64s still offer double the Conroe's L1 cache at 128 KB per core. That's a massive 256 KB of L1, while Conroe is Intel's largest L1 offering to the desktop segment at 64 KB per core. The L2 cache bus on the Conroe is 256 bits. The AMD 64's L2 cache is half as wide, at 128 bits; Intel still has double the L2 cache though (4 MB), and it's shared!